MANGANESE

(Data in thousand metric tons gross weight unless otherwise noted)

Domestic Production and Use: Manganese ore containing 20% or more manganese has not been produced domestically since 1970. Manganese ore was consumed mainly by eight firms with plants principally in the East and Midwest. Most ore consumption was related to steel production, either directly in pig iron manufacture or indirectly through upgrading the ore to ferroalloys. Additional quantities of ore were used for such nonmetallurgical purposes as production of dry cell batteries, in fertilizers and animal feed, and as a brick colorant. Manganese ferroalloys were produced at two plants. Construction, transportation, and machinery end uses accounted for about 37%, 15%, and 12%, respectively, of manganese consumption on a manganese-content basis. Most of the rest went to a variety of other iron and steel applications. In 2017, the value of domestic consumption, estimated from foreign trade data on a manganese-content basis, was about $940 million.

Salient Statistics—United States:[1]	2013	2014	2015	2016	2017[e]
Production, mine	—	—	—	—	—
Imports for consumption:					
Manganese ore	558	387	441	282	310
Ferromanganese	335	365	292	229	300
Silicomanganese[2]	329	448	301	264	370
Exports:					
Manganese ore	1	1	1	1	2
Ferromanganese	2	6	5	7	9
Silicomanganese	6	3	1	2	8
Shipments from Government stockpile excesses[3]					
Manganese ore	—	—	—	—	—
Ferromanganese	1	18	32	42	11
Consumption, reported:[4]					
Manganese ore	523	508	451	410	390
Ferromanganese	368	360	344	342	360
Silicomanganese	152	146	138	139	150
Consumption, apparent, manganese[5]	801	834	693	538	660
Price, average, 46% to 48% Mn metallurgical ore, dollars per metric ton unit, contained Mn:					
Cost, insurance, and freight (c.i.f.), U.S. ports[e]	4.61	4.49	3.53	3.41	4.40
China spot market (c.i.f.)	5.29	4.72	3.22	4.48	[6]5.88
Stocks, producer and consumer, yearend:[4]					
Manganese ore	217	189	187	207	200
Ferromanganese	27	23	21	21	22
Silicomanganese	6	10	21	10	10
Net import reliance[7] as a percentage of apparent consumption	100	100	100	100	100

Recycling: Manganese was recycled incidentally as a constituent of ferrous and nonferrous scrap; however, scrap recovery specifically for manganese was negligible. Manganese is recovered along with iron from steel slag.

Import Sources (2013–16): Manganese ore: Gabon, 73%; South Africa, 11%; Australia, 9%; Mexico, 3%; and other, 4%. Ferromanganese: South Africa, 51%; Australia, 15%; Norway, 12%; Republic of Korea, 11%; and other, 11%. Silicomanganese: Georgia, 31%; South Africa, 25%; Australia, 19%; Norway, 9%; and other, 16%. Manganese contained in principal manganese imports:[8] South Africa, 29%; Gabon, 22%; Australia, 14%; Georgia, 11%; and other, 24%.

Tariff: Item	Number	Normal Trade Relations 12–31–17
Ores and concentrates	2602.00.0040/60	Free.
Manganese dioxide	2820.10.0000	4.7% ad val.
High-carbon ferromanganese	7202.11.5000	1.5% ad val.
Ferrosilicon manganese (silicomanganese)	7202.30.0000	3.9% ad val.
Metal, unwrought	8111.00.4700/4900	14% ad val.

Depletion Allowance: 22% (Domestic), 14% (Foreign).

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MANGANESE

Government Stockpile:

Stockpile Status—9–30–17[9]

Material	Inventory	Authorized for disposal	Disposals FY 2017
Manganese ore, metallurgical grade	292	292	—
Ferromanganese, high-carbon	213	45	24

Events, Trends, and Issues: U.S. manganese apparent consumption was estimated to increase by 23% to 660,000 tons in 2017 compared with that in 2016. This was primarily a result of increases in U.S. ferromanganese and silicomanganese imports in response to the 4% increase in domestic steel production. The annual average domestic manganese ore contract price followed the 28% increase in the average U.S. import unit value of manganese ore imports containing 47% or more manganese. This increase in price reflected the decrease in manganese ore production worldwide during 2016 and the early part of 2017, particularly from several large producers in Australia, Gabon, and South Africa.

World Mine Production and Reserves (manganese content): Reserves for Australia, China, Gabon, Ghana, and India were revised based on data reported by the Governments of those countries.

	Mine production		Reserves[10]
	2016	2017[e]	
United States	—	—	—
Australia	2,240	2,200	[11]94,000
Brazil	1,080	1,200	120,000
China	2,330	2,500	48,000
Gabon	1,620	1,600	20,000
Ghana	553	550	13,000
India	745	790	34,000
Kazakhstan, concentrate	212	230	5,000
Malaysia	266	270	NA
Mexico	206	220	5,000
South Africa	5,300	5,300	200,000
Ukraine, concentrate	425	380	140,000
Other countries	681	760	Small
World total (rounded)	15,700	16,000	680,000

World Resources: Land-based manganese resources are large but irregularly distributed; those in the United States are very low grade and have potentially high extraction costs. South Africa accounts for about 78% of the world's identified manganese resources, and Ukraine accounts for about 10%.

Substitutes: Manganese has no satisfactory substitute in its major applications.

[e]Estimated. NA Not available. — Zero.
[1]Manganese content typically ranges from 35% to 54% for manganese ore and from 74% to 95% for ferromanganese.
[2]Imports more nearly represent amount consumed than does reported consumption.
[3]Defined as stockpile shipments – receipts. If receipts, a negative quantity is shown.
[4]Exclusive of ore consumed directly at iron and steel plants and associated yearend stocks.
[5]Defined as imports – exports + adjustments for Government and industry stock changes, thousand tons, manganese content. To avoid double counting, manganese consumption is not calculated as the sum of manganese ore, ferromanganese, and silicomanganese consumption because manganese in ore is used to produce ferromanganese and silicomanganese.
[6]Average weekly price through October 2017 for average metallurgical-grade ore containing 44% manganese, as reported by CRU Group.
[7]Defined as imports – exports + adjustments for Government and industry stock changes.
[8]Includes imports of ferromanganese, manganese ore, silicomanganese, synthetic manganese dioxide, and unwrought manganese metal.
[9]See Appendix B for definitions.
[10]See Appendix C for resource and reserve definitions and information concerning data sources.
[11]For Australia, Joint Ore Reserves Committee-compliant reserves were about 47 million tons of manganese content.